Virtualization Technology

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Hardware Virtualization in Cloud Computing

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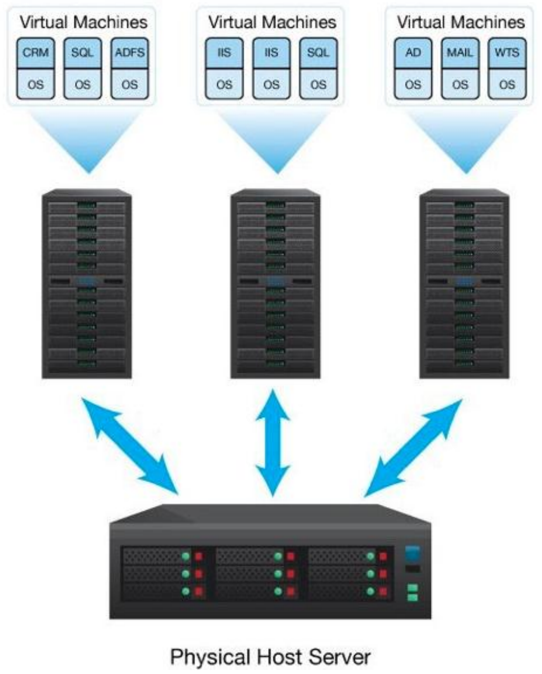
Virtualization

- Virtualization is the creation of a virtual (rather than actual) version of something, such as an operating system, server (hardware), storage device or network resources.
- Virtualization hides the physical characteristics from the users, presenting instead an abstract platform.

Refers to the practice of breaking down the physical infrastructure of computing and networking resources into smaller, reusable and more flexible units.

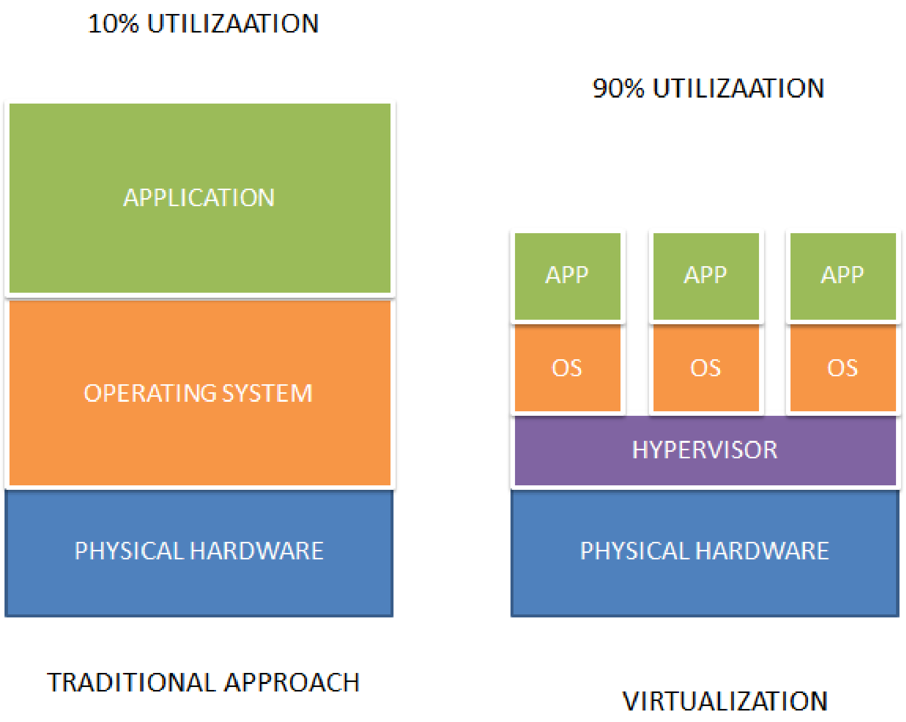


Hardware Virtualization : An Introduction

- Virtualization means creating a virtual platform of something, which will include virtual computer hardware, virtual storage devices, and virtual computer network.
- Hardware virtualization (server virtualization) is the abstraction of computing resources from the software that uses those resources.
- Using specially designed software's i.e. hypervisors multiple logical resources can be created within a single physical system.

Need for Hardware Virtualization

- Traditional approach Posed major faults for software configuration and made it difficult to move or reinstall software on different hardware, such as restoring backups after a fault or disaster.



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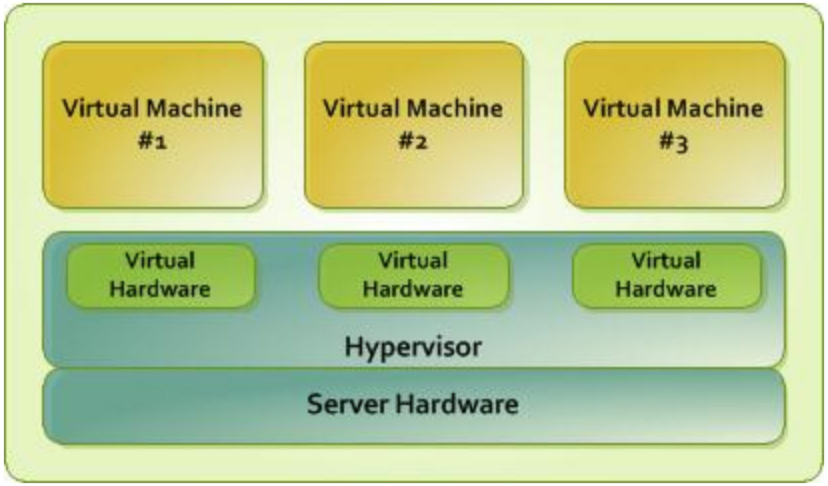
Working of Hardware Virtualization

- Hypervisor creates an abstraction layer between the software and the hardware in use.
- After the installation of a hypervisor, virtual representations take place such as virtual processors.
- Virtualized computing resources are provisioned into isolated instances called *VMs*, where OSes and applications can be installed.



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Types of Hardware Virtualization

- Full Virtualization
  - Software Assisted Virtualization
  - Hardware Assisted Virtualization
- Paravirtualization
- Partial virtualization

Full Virtualization

- Introduced by IBM in the year 1966.
- Guest OS is completely isolated by the VM from the virtualization layer and hardware.
- Refers to the ability of running a program, most likely an OS, on top of a VM directly and without any modification.
- Seems as if the program is running on the raw hardware.
- VMM are required to provide complete emulation of the entire underlying hardware.

Advantages

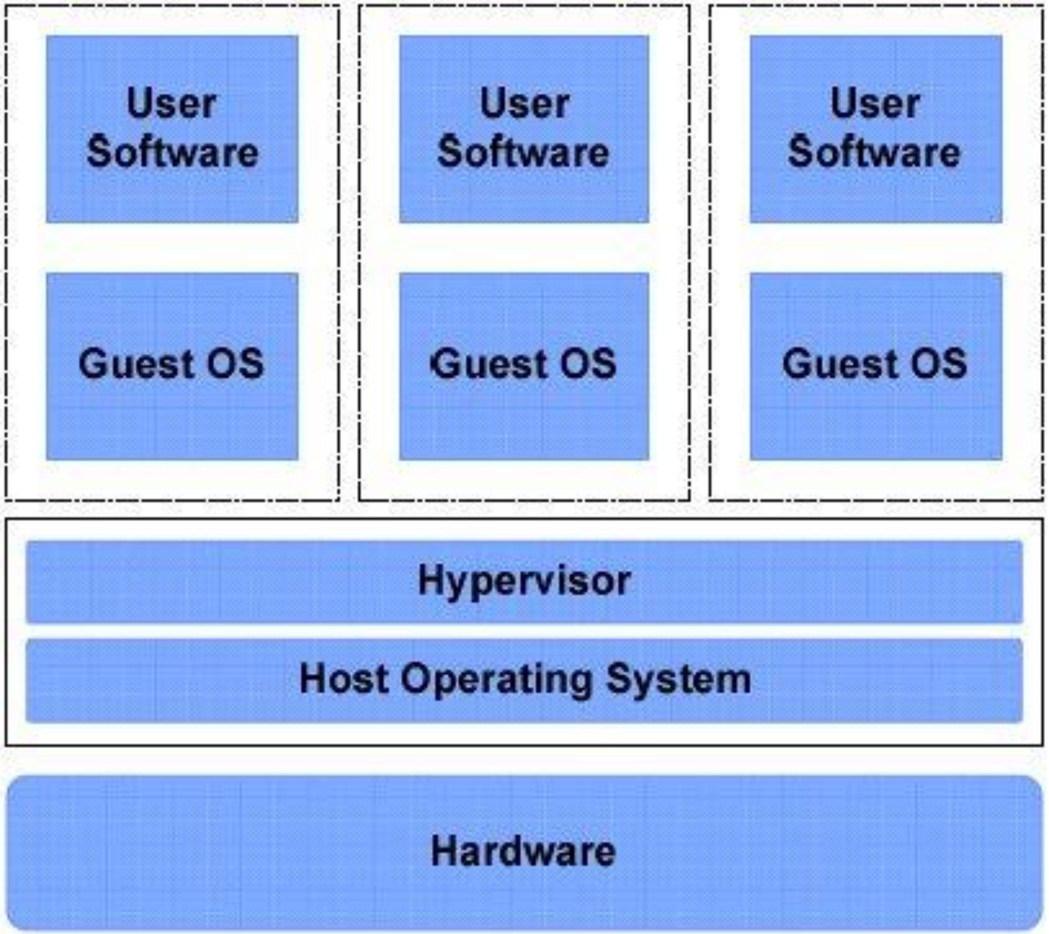
- Enhanced Security (Due to Complete Isolation)
- Ease of Emulation of Different Architectures
- Coexistence of Different Systems on the same Platform

Challenges

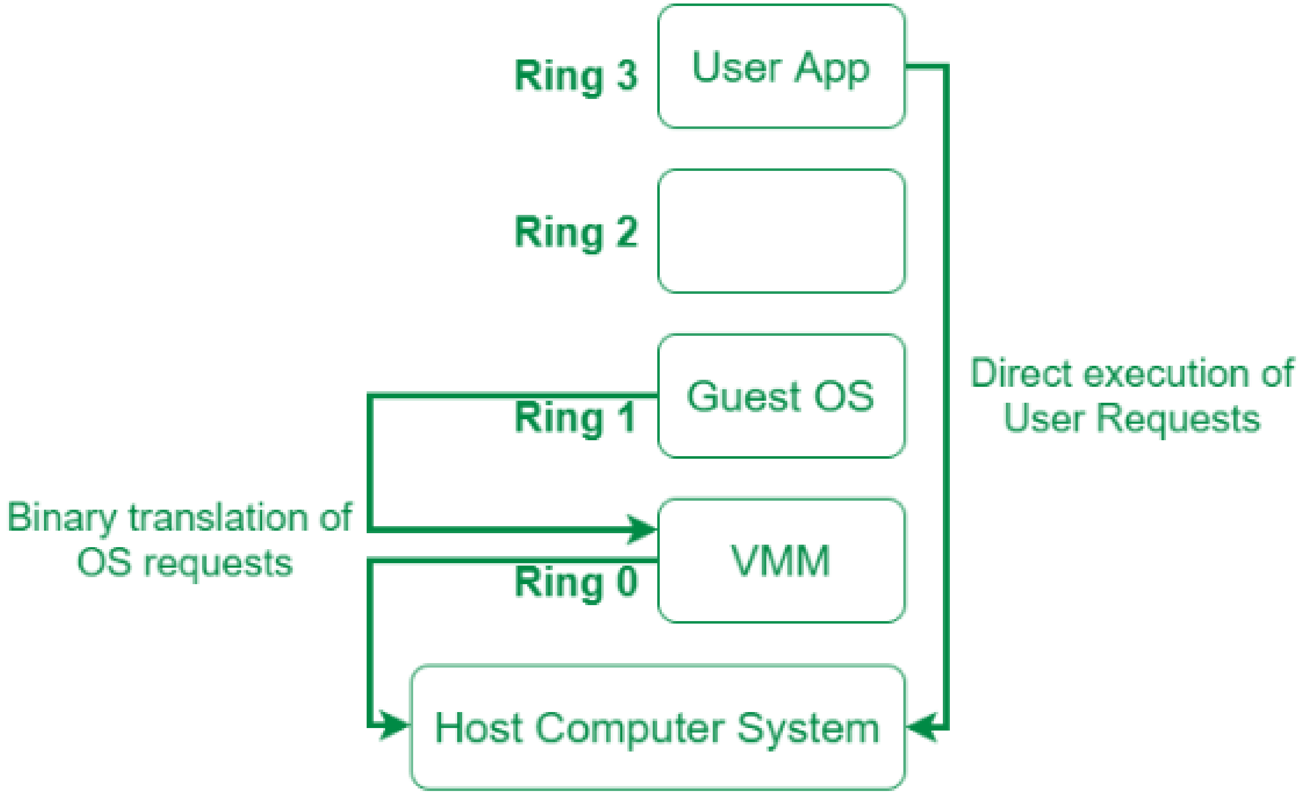
- Interception of privileged instruction (IO instructions)

Full Virtualization

Full Virtualization



Full Virtualization



Software Assisted Full Virtualization

- Software-assisted full virtualization typically involves time consuming processes such as binary translation.
- Binary translation virtualization is implemented through the trap-and-emulate approach
- The hypervisor configures the CPU so that all potentially unsafe instructions will cause a trap, or an exceptional condition that transfers control back to the hypervisor.
- Steps the hypervisor takes if it receives a trap :
    - Inspects the instruction
    - Emulates it in a safe way
    - Continues execution of the instruction.
- **Due to binary translation, it is often criticized for performance issue.**

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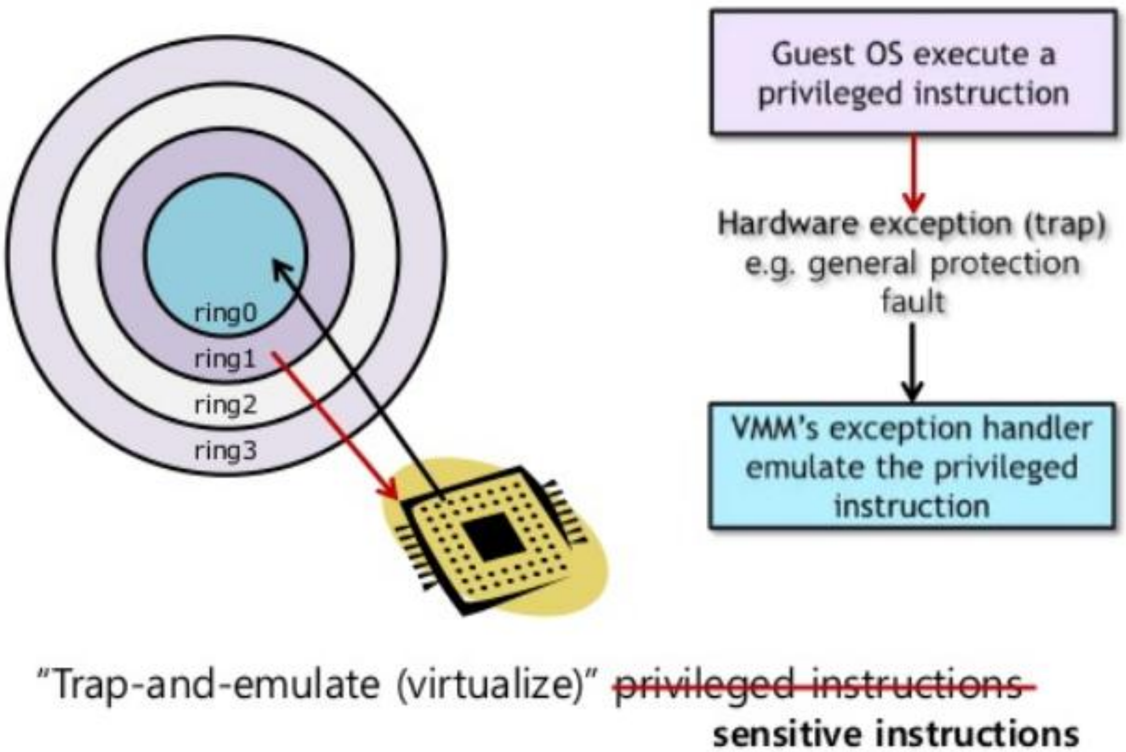
Software Assisted Full Virtualization

Trap and Emulation



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Sensitive Instructions

- **Class of Instructions**

Normal instructions

Decided by architecture

- Not trapped by privilege layer

Privileged instructions

- Automatically trapped by privilege layer

Sensitive instructions

- Must be emulated (virtualized) for fidelity and safety
- e.g., Processor mode changes, HW accesses, …

Decided by VMM

- **Virtualizable Architecture**
  - Sensitive Instructions ⊆ Privileged Instructions
  - Trap-and-Emulate every sensitive instruction

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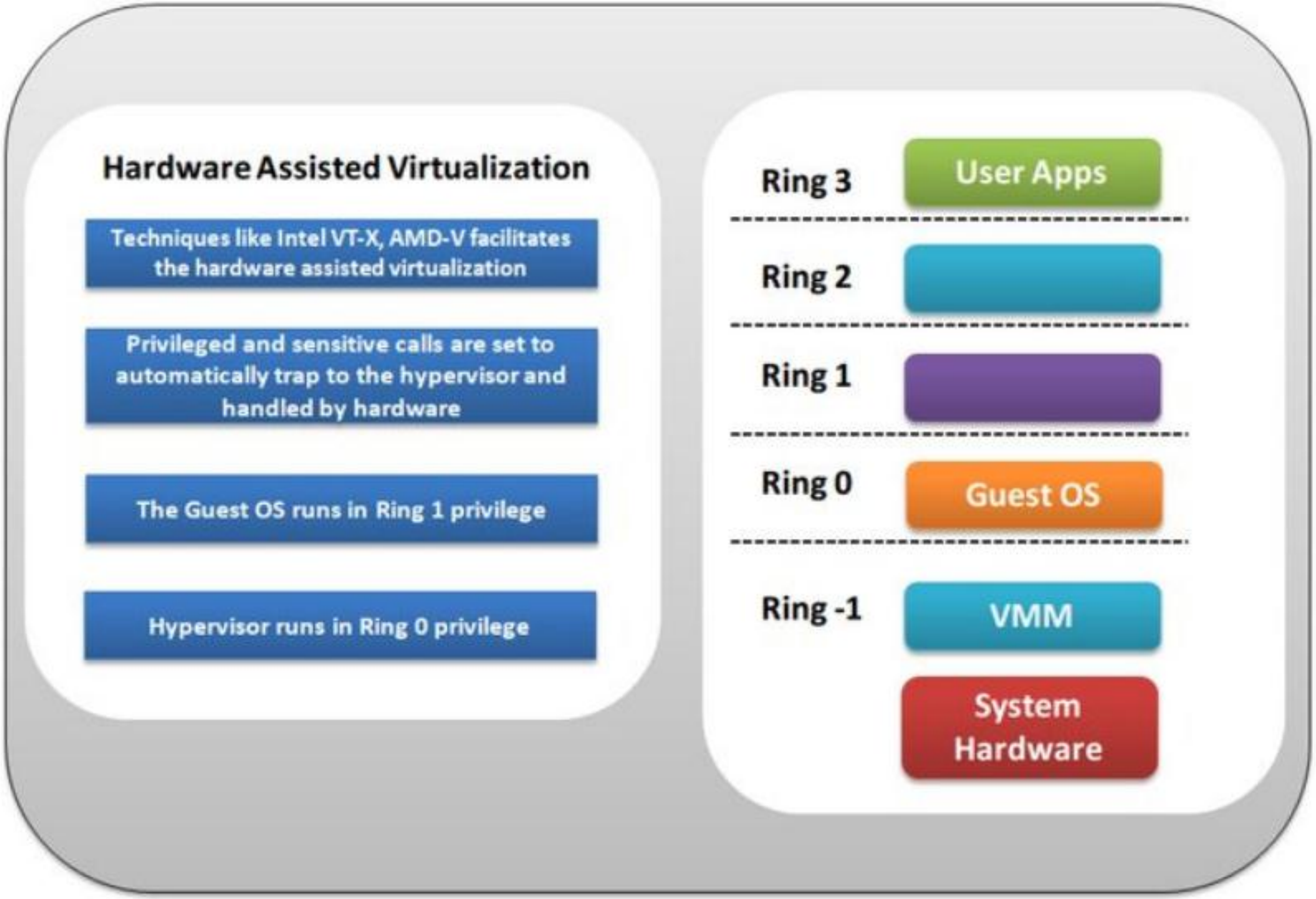
Hardware Assisted Full Virtualization

- Originally introduced in the IBM/370
- Eliminates the binary translation (Directly interrupts with hardware using the virtualization technology)
- Virtualization approach that enables efficient full virtualization using the help from hardware capabilities (Has been integrated on X86 processors since 2005 (Intel VT-x and AMD-V)).
- Guest OS's instructions might allow a virtual context execute privileged instructions directly on the processor, even though it is virtualized.
- Examples of hardware assisted virtualization :
  - KVM, Virtual Box, Xen, VMware

Hardware Assisted Virtualization

- No binary translation of hardware instructions
- Commands are executed (directed) towards the hardware through the hypervisor.
- The VMM can handle the sensitive instructions using a classic trap-and-emulate model in hardware, as opposed to software.
- Hypervisors supporting this technology can load at Ring -1 and guest OSes can access the CPU at Ring 0 (as normally done when running on a physical host).
- Allows for guest OS to be virtualized without any modifications.

Hardware Assisted Virtualization

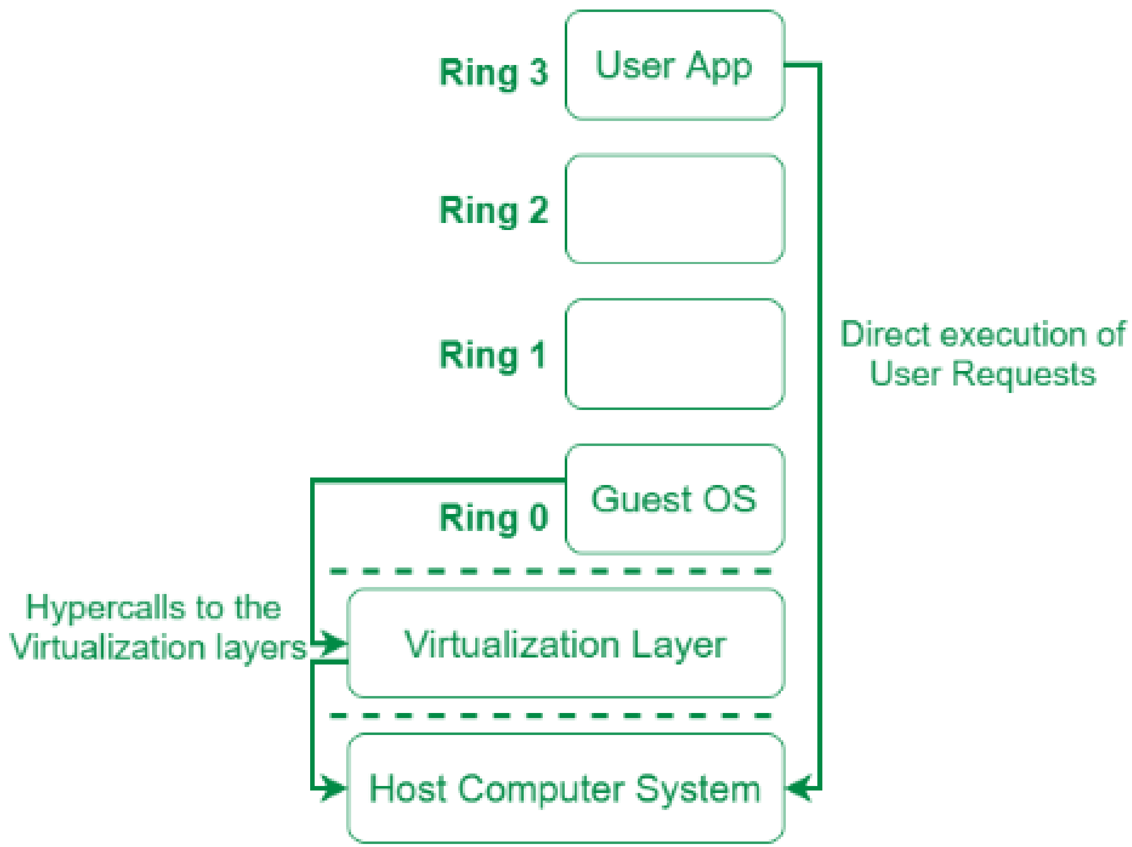


Paravirtualization

- Guest OS is recompiled, installed inside a VM, and operated on top of a hypervisor program running on the host OS.
- Virtualization method modifies the guest operating system (recompiled) to communicate with the hypervisor.
- Developed to provide better performance than full virtualization by permitting the guest OS to recognize the presence of the hypervisor and communicate with the hypervisor directly (hyper calls).
- Unlike full virtualization, it does not need to emulate hardware for VMs.
- Instead, it provisions an interface to VMs which is somewhat similar to the underlying hardware.

Types of Hardware Virtualization

Paravirtualization



Types of Hardware Virtualization

Uses of Paravirtualization

- Disaster recovery
- Capacity management
- Separating test systems and development environments
- Transferring data from one system to another

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Types of Hardware Virtualization

Advantages of Paravirtualization

- Easier Backups
- Fast Migrations
- Improved System Utilization
- Server Consolidation
- Power Conservation

Disadvantages of Paravirtualization

- Requirement to modify the kernel of the guest OS before installation
- It is not possible to modify the kernel of proprietary OS i.e. Windows
- Compelled to use open source software's

Types of Hardware Virtualization

Full Virtualization	Paravirtualization
VM permit the execution of the instructions with running of unmodified Guest OS in an entire isolated way.	VM does not implement full isolation of OS but rather provides a different API which is utilized when OS is subjected to alteration.
Less secure.	More secure.
uses binary translation and direct approach as a technique for operations.	Uses hypercalls at compile time for operations
Slow	Fast
More portable and compatible.	Less portable and compatible.
Example: Microsoft and Parallels systems.	Example : VMware and Xen

Thank You